

Haripriya Varanasi

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Education

George Mason University

Master of Science in Data Analytics Engineering, GPA: 3.73/4.0

Fairfax, VA

Jan 2024 – May 2026

Andhra University

Bachelor of Technology in Computer Science Engineering, GPA: 8.22/10

Visakhapatnam, India

Jan 2020 – May 2024

Professional Experience

Blue Horse Digital

AI Data Solutions Analyst (Volunteer)

Jul 2026 – Present

Remote

- Prepare and analyze datasets for software applications and business projects through data collection, cleaning, transformation, and validation.
- Support the development of machine learning models and reproducible data analysis pipelines to generate reliable analytical outputs.
- Collaborate with front-end and back-end teams to integrate data systems and test, debug, and troubleshoot data-related application issues.

George Mason University

Graduate Research and Teaching Assistant

May 2025 – May 2026

Fairfax, VA

- Reviewed and assessed 40+ graduate-level analytics projects involving data preprocessing, feature engineering, machine learning, and model evaluation.
- Guided students through assessments and technical problem-solving, resolving questions involving Python, data analysis, machine learning workflows, debugging, and model interpretation.
- Developed reproducible evaluation guidelines and technical documentation to standardize project reviews, data quality checks, feedback, and assessment procedures.

Gulfstream Foundation

Data Analyst / Scrum Master Intern

Jan 2026 – May 2026

Fairfax, VA

- Led analytics delivery for a 6-person agile team building an offline flood risk decision-support platform for Puerto Rico's 78 municipalities.
- Designed a 5-stage DuckDB pipeline integrating 10 federal sources, including FEMA, USGS, NWS, NOAA, and ACS Census, into hazard, vulnerability, and response-readiness indicators.
- Delivered a 5-tab Streamlit dashboard with risk maps, rankings, live conditions, chatbot-style queries, and SQL exploration; presented and demoed the platform at STAR-TIDES 2026.

BrainOVision

AWS Cloud Intern

Jan 2023 – Jul 2023

Hyderabad, India

- Completed hands-on AWS cloud training and project work using S3, EC2, IAM, Lambda, and RDS.
- Supported cloud-based application and data workflows involving storage, compute provisioning, access control, database connectivity, Python, SQL, and basic ETL.

Personal Projects

CFPB Consumer Financial Complaint Analysis

Python, DuckDB, Pandas, Scikit-learn, Plotly, Streamlit – 16M+ Records

Jun 2026

- Built a multi-tab DuckDB analytics dashboard querying 16M+ CFPB complaint records, including a weighted Company Accountability Index and rolling z-score anomaly detection for complaint spikes.
- Developed an interactive state-level choropleth of relief rates and a Random Forest relief-prediction model using a 200K-row sample, achieving 72% accuracy.

US Air Quality & Community Health Risk Platform

Python, AWS S3, GitHub Actions, Streamlit, EPA AQI API, CDC PLACES API, US Census API – 3,000+ Counties

Apr 2026

- Integrated EPA, CDC, and Census API data across 3,000+ counties into a standardized master dataset.
- Built automated ingestion and validation workflows with GitHub Actions and AWS S3 for county-level analytics.

SmartPave: Predictive Road Deterioration Modeling

PySpark, Apache Spark, Databricks, XGBoost, Random Forest

May 2025

- Built a large-scale ML pipeline on Databricks to predict road deterioration from sensor and inspection data.
- Applied hyperparameter tuning and cross-validation across XGBoost and Random Forest models to support maintenance planning.

Health Risk Analytics: Classification and Clustering

Python, Scikit-learn, Logistic Regression, Random Forest, K-Means, PCA

Oct 2024

- Applied K-Means and PCA to segment patient populations and surface high-risk groups; resolved class imbalance via upsampling and weighted models; evaluated classifiers across accuracy, precision, recall, and F1.

Technical Skills

Languages/Query: Python, R, SQL, PySpark

Analysis/BI: Excel, Pandas, NumPy, Scikit-learn, ggplot2, Matplotlib, Plotly, Streamlit, Power BI, Tableau

Cloud/Data Platforms: AWS S3, EC2, IAM, Lambda, RDS, Databricks, DuckDB, Apache Spark, Git, GitHub, GitHub Actions, Jupyter Notebook

Core Competencies: Data Cleaning, ETL Pipelines, REST API Integration, Machine Learning, Feature Engineering, Statistical Modeling, Model Evaluation, Data Validation, Requirements Analysis, Stakeholder Reporting, Agile/Scrum